

Program1:Creating “Hello world” Application.

Procedure:

1. Open Android Studio and create a new project.
2. Choose 'Empty Views Activity' and name your project.

`activity\_main.xml` code:

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
xmlns:app="http://schemas.android.com/apk/res-auto"
xmlns:tools="http://schemas.android.com/tools"
android:id="@+id/main"
android:layout_width="match_parent"
android:layout_height="match_parent"
tools:context=".MainActivity">

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Hello World!"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent" />

</androidx.constraintlayout.widget.ConstraintLayout>
```

`MainActivity.java`

```
package com.example.helloworld;

import android.os.Bundle;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {
    @Override
    protected void onCreate(Bundle savedInstanceState) {
```

```
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
    }
}
```

Output:

The application will launch and display the text 'Hello World!' centered on the screen.

Program 2: Creating an application that displays message based on the screen orientation.**Procedure:**

Open Android Studio and create a new Empty Views Activity project.

MainActivity.java Code:

```
package com.example.orientation;
```

```
import android.content.res.Configuration;
```

```
import android.os.Bundle;
```

```
import android.widget.TextView;
```

```
import androidx.activity.EdgeToEdge;
```

```
import androidx.appcompat.app.AppCompatActivity;
```

```
import androidx.core.graphics.Insets;
```

```
import androidx.core.view.ViewCompat;
```

```
import androidx.core.view.WindowInsetsCompat;
```

```
public class MainActivity extends AppCompatActivity {
```

```
    @Override
```

```
    protected void onCreate(Bundle savedInstanceState) {
```

```
        super.onCreate(savedInstanceState);
```

```
        EdgeToEdge.enable(this);
```

```
        setContentView(R.layout.activity_main);
```

```
        ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main), (v, insets) -> {
```

```
            Insets systemBars = insets.getInsets(WindowInsetsCompat.Type.systemBars());
```

```

        v.setPadding(systemBars.left, systemBars.top, systemBars.right,
systemBars.bottom);
        return insets;
    });

    TextView t = findViewById(R.id.tid);
    if (getResources().getConfiguration().orientation ==
Configuration.ORIENTATION_LANDSCAPE) {
        t.setText("Landscape Mode");
    } else {
        t.setText("Portrait Mode");
    }
}
}

```

activity_main.xml Code:

```

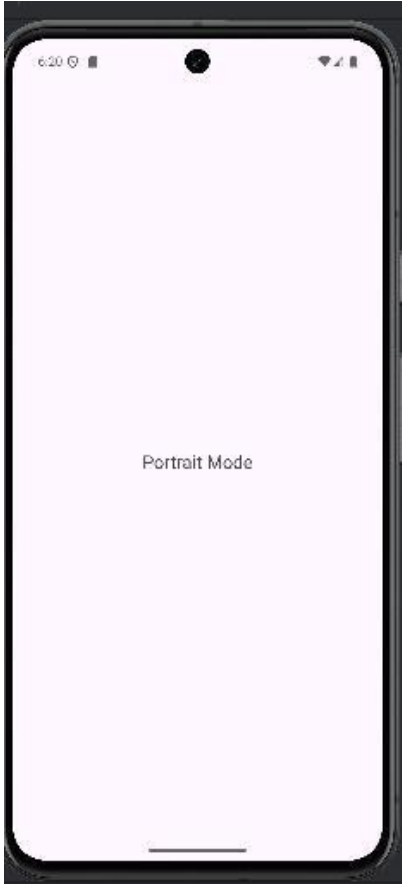
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
xmlns:app="http://schemas.android.com/apk/res-auto"
xmlns:tools="http://schemas.android.com/tools"
android:id="@+id/main"
android:layout_width="match_parent"
android:layout_height="match_parent"
tools:context=".MainActivity">

    <TextView
        android:id="@+id/tid"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Orientation"
        android:textSize="20sp"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent" />

```

```
</androidx.constraintlayout.widget.ConstraintLayout>
```

Output:



Program 3: Create an application to develop Login window using UI controls.

Procedure:

1. Open Android Studio and create a new project.
2. Choose an Empty Views Activity and name it MainActivity.
3. Open activity_main.xml.
4. Add two EditText components for username and password.
5. Add one Button component labeled "Login".
6. Use proper layout structure (e.g., LinearLayout or ConstraintLayout).
7. Set hints and input types for EditText.
8. Apply necessary styling like padding, margin, and alignment.
9. Run the app to verify the layout visually.

XML Layout (activity_main.xml):

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="24dp"
    android:gravity="center"
    android:background="#FFFFFF">

    <EditText
        android:id="@+id/username"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Username"
        android:inputType="textPersonName" />

    <EditText
        android:id="@+id/password"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Password"
        android:inputType="textPassword"
        android:layout_marginTop="12dp" />

    <Button
        android:id="@+id/loginButton"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Login"
        android:layout_marginTop="18dp" />
</LinearLayout>
```

Program 4: Create a sample application with login module(check user name and password) On successful login change Textview “Login Successful”. On login fail alert using Toast “login fail”

Procedure:

1. Open Android Studio and create a New Project named EmailApp.
2. Choose Empty Views Activity and name the main activity as MainActivity.
3. Open activity_main.xml and design the layout using EditText for username and password, and a Button for login.
4. In MainActivity.java, write code to handle button click and validate user input.
5. Run the app on an emulator or physical device.
6. Enter various combinations of usernames and passwords to check the validation logic.

Source Code:

MainActivity.java

```
package com.example.emailapp;

import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.Toast;

import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_main);
        ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main), (v, insets) -> {
            Insets systemBars = insets.getInsets(WindowInsetsCompat.Type.systemBars());
            v.setPadding(systemBars.left, systemBars.top, systemBars.right,
```

```

systemBars.bottom);
    return insets;
});

EditText et1 = findViewById(R.id.usr);
EditText et2 = findViewById(R.id.pwd);
Button buttonLogin = findViewById(R.id.btn);

buttonLogin.setOnClickListener(new View.OnClickListener() {
    @Override
    public void onClick(View v) {
        String username = et1.getText().toString().trim();
        String password = et2.getText().toString().trim();

        if (username.isEmpty() || password.isEmpty()) {
            Toast.makeText(MainActivity.this, "Username or Password is empty",
Toast.LENGTH_SHORT).show();
        } else if (username.equals("admin") && password.equals("welcome")) {
            Toast.makeText(MainActivity.this, "Login Successful",
Toast.LENGTH_SHORT).show();
        } else {
            Toast.makeText(MainActivity.this, "Incorrect username or password",
Toast.LENGTH_SHORT).show();
        }
    }
});
}
}

```

activity_main.xml

```

<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="24dp"
    android:gravity="center"
    android:background="#EFEFEF">

```

```
<EditText
    android:id="@+id/usr"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:hint="Enter Username"
    android:inputType="textPersonName" />
```

```
<EditText
    android:id="@+id/pwd"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:hint="Enter Password"
    android:inputType="textPassword"
    android:layout_marginTop="12dp" />
```

```
<Button
    android:id="@+id/btn"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Login"
    android:layout_marginTop="18dp"
    android:backgroundTint="@android:color/holo_blue_dark"
    android:textColor="#FFFFFF"/>
```

```
</LinearLayout>
```

Output:

Input: Empty username/password → Output: “Username or Password is empty”

Input: admin / welcome → Output: “Login Successful”

Input: any other values → Output: “Incorrect username or password”

Program 5: Create an application to implement new activity using explicit intent, implicit intent and content provider.

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
```



```

<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">
    <Button
        android:id="@+id/btn1"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_marginTop="36dp"
        android:text="Explicit Intent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.365"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent" />
    <Button
        android:id="@+id/btn2"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Implicit Content"
        app:layout_constraintBottom_toTopOf="@+id/btn3"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.376"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toBottomOf="@+id/btn1"
        app:layout_constraintVertical_bias="0.471" />
    <Button
        android:id="@+id/btn3"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_marginBottom="468dp"
        android:text="Open Contacts"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.376"
        app:layout_constraintStart_toStartOf="parent" />
</androidx.constraintlayout.widget.ConstraintLayout>

```

MainActivity.java

```
package com.example.intent;
```

```

import android.content.Intent;
import android.net.Uri;
import android.os.Bundle;
import android.provider.ContactsContract;
import android.view.View;
import android.widget.Button;

import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_main);
        ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main), (v,
insets) -> {
            Insets systemBars = insets.getInsets(WindowInsetsCompat.Type.systemBars());
            v.setPadding(systemBars.left, systemBars.top, systemBars.right,
systemBars.bottom);
            return insets;
        });
        Button b1=findViewById(R.id.btn1);
        Button b2=findViewById(R.id.btn2);
        Button b3=findViewById(R.id.btn3);
        b1.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View view) {
                Intent exintent=new Intent(MainActivity.this, MainActivity2.class);
                startActivity(exintent);
            }
        });
        b2.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View view) {

```

```

        Intent impintent=new Intent(Intent.ACTION_VIEW,
Uri.parse("https://www.google.com"));
        startActivity(impintent);
    }
});
b3.setOnClickListener(new View.OnClickListener() {
    @Override
    public void onClick(View view) {
        Intent intent = new Intent(Intent.ACTION_PICK,
ContactsContract.Contacts.CONTENT_URI);
        startActivity(intent);
    }
});
}
}

```

Steps to Create Activity 2 in Android Studio

Steps to Create Activity 2

1. 1. Right-click on the `java` folder in `app > java > your\_package\_name`.
2. 2. Select New > Activity > Empty view Activity.
3. 3. Enter the Activity Name as `MainActivity2`.
4. 4. Keep the layout name as `activity\_main2.xml` (or rename as needed).
5. 5. Click Finish. This creates the Java class, layout XML, and updates the manifest.

Java Code for MainActivity2.java

```

package com.example.intent;

import android.os.Bundle;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity2 extends AppCompatActivity {
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main2); // Refers to XML file
    }
}

```

```
}
```

XML Layout for activity_main2.xml

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity2">

    <Button
        android:id="@+id/button"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Second Activity"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.564"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent"
        app:layout_constraintVertical_bias="0.516" />
</androidx.constraintlayout.widget.ConstraintLayout>
```

Program 6: Creating an UI with All Views

activity_main.xml Code:

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
```

```
tools:context=".MainActivity">
```

```
<LinearLayout  
    android:layout_width="match_parent"  
    android:layout_height="match_parent"  
    android:orientation="vertical"  
    android:textAlignment="center">
```

```
<TextView  
    android:id="@+id/textView3"  
    android:layout_width="match_parent"  
    android:layout_height="wrap_content"  
    android:text="WELCOME"  
    android:textAlignment="center" />
```

```
<EditText  
    android:id="@+id/editTextText3"  
    android:layout_width="match_parent"  
    android:layout_height="wrap_content"  
    android:ems="10"  
    android:inputType="text"  
    android:text="Enter your name here" />
```

```
<RadioGroup  
    android:layout_width="match_parent"  
    android:layout_height="match_parent" >
```

```
<RadioButton  
    android:id="@+id/radioButton"  
    android:layout_width="match_parent"  
    android:layout_height="wrap_content"  
    android:text="Male" />
```

```
<RadioButton  
    android:id="@+id/radioButton2"  
    android:layout_width="match_parent"  
    android:layout_height="wrap_content"  
    android:text="Female" />
```

```
<TextView  
    android:id="@+id/textView5"  
    android:layout_width="match_parent"  
    android:layout_height="wrap_content"  
    android:text="Languages Known" />
```

```
<CheckBox
```

```
android:layout_width="match_parent"  
android:layout_height="wrap_content"  
android:text="English" />
```

```
<CheckBox  
    android:id="@+id/checkBox2"  
    android:layout_width="match_parent"  
    android:layout_height="wrap_content"  
    android:text="Kannada" />
```

```
<ProgressBar  
    android:id="@+id/progressBar"  
    style="?android:attr/progressBarStyle"  
    android:layout_width="match_parent"  
    android:layout_height="wrap_content" />
```

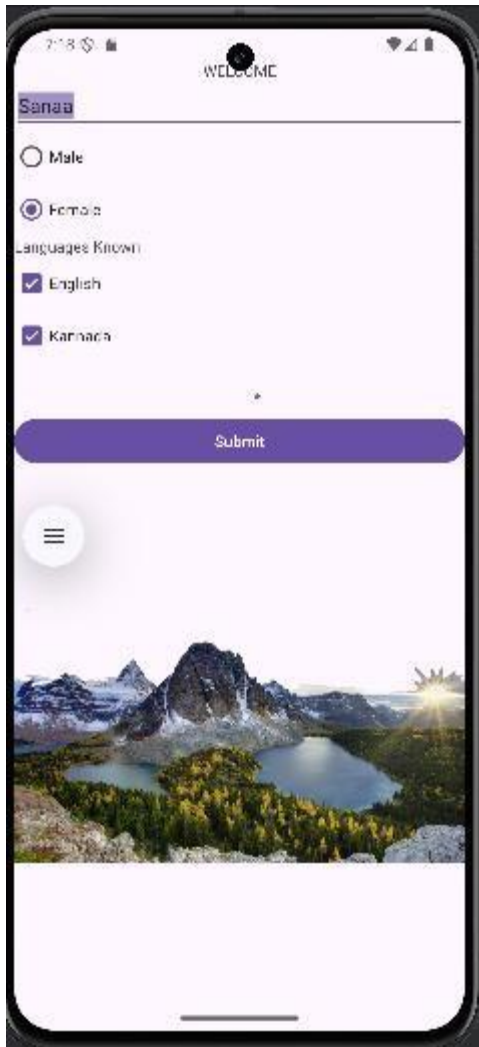
```
<Button  
    android:id="@+id/button4"  
    android:layout_width="match_parent"  
    android:layout_height="wrap_content"  
    android:text="Submit" />
```

```
<ImageView  
    android:id="@+id/imageView2"  
    android:layout_width="match_parent"  
    android:layout_height="wrap_content"  
    app:srcCompat="@drawable/img" />
```

```
</RadioGroup>
```

```
</LinearLayout>  
</androidx.constraintlayout.widget.ConstraintLayout>
```

Output:



Program 7:Creating Menu in Application

Step 1: Create menu folder

1. Go to the 'res' directory in Android Studio.
2. Right-click on 'res' > New > Android Resource Directory.
3. In the 'Resource type' dropdown, select 'menu'.
4. Click 'OK'.

Step 2: Create menu_main.xml file

1. Right-click on the newly created 'menu' folder.
2. Select New > Menu Resource File.
3. Name the file as menu_main.xml.

4. Click 'OK'.

Step 3: Add menu items to menu_main.xml

write the following code inside menu_main.xml:

```
<menu xmlns:android="http://schemas.android.com/apk/res/android">
    <item
        android:id="@+id/action_search"
        android:title="Search"
        android:showAsAction="ifRoom" />
    <item
        android:id="@+id/action_settings"
        android:title="Settings"
        android:showAsAction="ifRoom" />
</menu>
```

Step 4: Add Toolbar using Palette

1. Open activity_main.xml.
2. Go to the 'Palette' on the left.
3. Under 'Widgets', drag and drop 'Toolbar' into your layout.
4. Give it an ID like: android:id="@+id/toolbar".

Step 5: Java Code for Toolbar and Menu

Use the following code in MainActivity.java:

```
package com.example.actionbarr;

import android.os.Bundle;
import android.widget.Toast;
import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.appcompat.widget.Toolbar;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;
```



```

public class MainActivity extends AppCompatActivity {
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_main);

        ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main), (v,
insets) -> {
            Insets systemBars = insets.getInsets(WindowInsetsCompat.Type.systemBars());
            v.setPadding(systemBars.left, systemBars.top, systemBars.right,
systemBars.bottom);
            return insets;
        });

        // Initialize and set the toolbar
        Toolbar toolbar = findViewById(R.id.toolbar);
        setSupportActionBar(toolbar);
    }

    // Menu creation
    @Override
    public boolean onCreateOptionsMenu(android.view.Menu menu) {
        getMenuInflater().inflate(R.menu.menu_main, menu);
        return true;
    }

    // Menu item click actions
    @Override
    public boolean onOptionsItemSelected(android.view.MenuItem item) {
        int id = item.getItemId();
        if (id == R.id.action_search) {
            Toast.makeText(this, "Search clicked", Toast.LENGTH_SHORT).show();
            return true;
        } else if (id == R.id.action_settings) {
            Toast.makeText(this, "Settings clicked", Toast.LENGTH_SHORT).show();
            return true;
        }
        return super.onOptionsItemSelected(item);
    }
}

```

```
}
```

Output:



Program 8:Create an Application with Custom Designed Opening Screen

Android Lab Manual – Splash Screen using Explicit Intent

1. Create Two Activities

- SplashActivity (First screen shown when the app launches)
- MainActivity (main content screen)

2. spashActivity.java

```
public class spashActivity extends AppCompatActivity {
```

```
    @Override
```

```
    protected void onCreate(Bundle savedInstanceState) {
```

```
        super.onCreate(savedInstanceState);
```

```
        EdgeToEdge.enable(this);
```

```

        setContentView(R.layout.activity_splash);
        ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main), (v,
insets) -> {
            Insets systemBars = insets.getInsets(WindowInsetsCompat.Type.systemBars());
            v.setPadding(systemBars.left, systemBars.top, systemBars.right,
systemBars.bottom);
            return insets;
        });
        new Handler().postDelayed(new Runnable() {
            @Override
            public void run() {
                Intent intent = new Intent(splashActivity.this, MainActivity.class);
startActivity(intent);
                finish();
            }
        },3000);
    }
}

```

3. activity_splash.xml (Splash Screen Layout with ConstraintLayout)

```

<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
xmlns:app="http://schemas.android.com/apk/res-auto"
xmlns:tools="http://schemas.android.com/tools"
android:id="@+id/main"
android:layout_width="match_parent"
android:layout_height="match_parent"
tools:context=".splashActivity">

```

```

    <TextView
        android:id="@+id/textView"
        android:layout_width="172dp"
        android:layout_height="20dp"
        android:layout_marginBottom="176dp"
        android:text="Welcome to app"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.389"
        app:layout_constraintStart_toStartOf="parent" />

```

```

<ImageView
    android:id="@+id/imageView2"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_marginStart="16dp"
    app:layout_constraintBottom_toTopOf="@+id/textView"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toTopOf="parent"
    app:srcCompat="@drawable/img" />

</androidx.constraintlayout.widget.ConstraintLayout>

```

4. MainActivity.java

```

public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
    }
}

```

5. AndroidManifest.xml

```

<manifest xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools">

    <application
        android:allowBackup="true"
        android:dataExtractionRules="@xml/data_extraction_rules"
        android:fullBackupContent="@xml/backup_rules"
        android:icon="@mipmap/ic_launcher"
        android:label="@string/app_name"
        android:roundIcon="@mipmap/ic_launcher_round"
        android:supportRtl="true"
        android:theme="@style/Theme.SpashScreenApp"
        tools:targetApi="31">

```

```

<activity
    android:name=".MainActivity"
    android:exported="false" />
<activity
    android:name=".spashActivity"
    android:exported="true">
    <intent-filter>
        <action android:name="android.intent.action.MAIN" />
        <category android:name="android.intent.category.LAUNCHER" />
    </intent-filter>
</activity>
</application>

</manifest>

```

Program 9:Read/Write the Local data

activity_main.xml Code:

```

<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

```

```

<LinearLayout
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical">

```

```

<EditText
    android:id="@+id/editName"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:ems="10"
    android:inputType="text"
    android:text="Enter Name" />

```

```

<Button
    android:id="@+id/btnSave"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="Save Preferences" />

<TextView
    android:id="@+id/txtResult"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="Hello" />

</LinearLayout>

</androidx.constraintlayout.widget.ConstraintLayout>

```

[MainActivity.java:](#)

```

package com.example.userprefapp;

import android.content.SharedPreferences;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.TextView;

import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_main);
    }
}

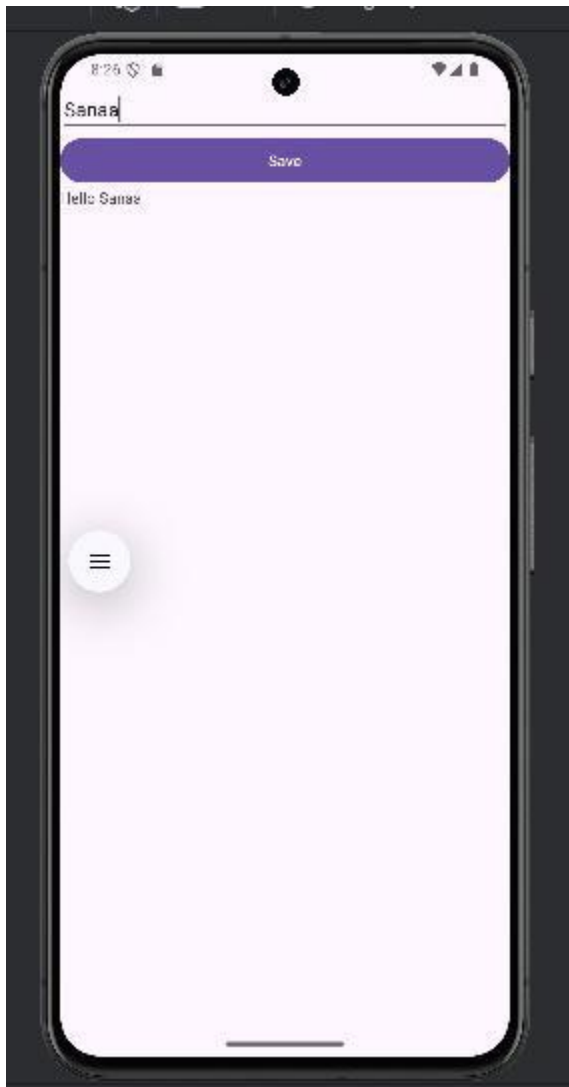
```

```

        ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main), (v, insets)
-> {
            Insets systemBars = insets.getInsets(WindowInsetsCompat.Type.systemBars());
            v.setPadding(systemBars.left, systemBars.top, systemBars.right,
systemBars.bottom);
            return insets;
        });
        EditText editName = findViewById(R.id.editName);
        Button btnSave = findViewById(R.id.btnSave);
        TextView txtResult = findViewById(R.id.txtResult);
        SharedPreferences prefs = getPreferences(MODE_PRIVATE);
        txtResult.setText("Hello " + prefs.getString("name", ""));
        btnSave.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View view) {
                prefs.edit().putString("name", editName.getText().toString()).apply();
                txtResult.setText("Hello " + editName.getText().toString());
            }
        });
    }
}

```

Output:



Program 10: Create / Read / Write data with database (SQLite).

activity_main.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
xmlns:app="http://schemas.android.com/apk/res-auto"
xmlns:tools="http://schemas.android.com/tools"
android:id="@+id/main"
android:layout_width="match_parent"
android:layout_height="match_parent"
tools:context=".MainActivity">

<LinearLayout
xmlns:android="http://schemas.android.com/apk/res/android"
android:orientation="vertical"
android:padding="16dp"
android:layout_width="match_parent"
android:layout_height="match_parent">

<EditText android:id="@+id/input"
android:layout_width="match_parent"
android:layout_height="wrap_content"
android:hint="Enter name" />

<Button android:id="@+id/save"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:text="Save" />

<TextView android:id="@+id/output"
android:layout_width="match_parent"
android:layout_height="wrap_content" />
```

```
</LinearLayout>
</androidx.constraintlayout.widget.ConstraintLayout>
```

[MainActivity.java](#)

```
package com.example.sqlldbapp;

import android.database.Cursor;
import android.database.sqlite.SQLiteDatabase;
import android.os.Bundle;
import android.widget.Button;
import android.widget.EditText;
import android.widget.TextView;

import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

public class MainActivity extends AppCompatActivity {
    TextView output;
    SQLiteDatabase db;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_main);
        ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main), (v, insets)
-> {
            Insets systemBars = insets.getInsets(WindowInsetsCompat.Type.systemBars());
            v.setPadding(systemBars.left, systemBars.top, systemBars.right,
systemBars.bottom);
            return insets;
        });

        EditText input = findViewById(R.id.input);
        output= findViewById(R.id.output);
        Button save = findViewById(R.id.save);

        // Create/open database and table
```

```

db = openOrCreateDatabase("MyDB", MODE_PRIVATE, null);
db.execSQL("CREATE TABLE IF NOT EXISTS student(name TEXT)");

// Save name to database
save.setOnClickListener(v -> {
    String name = input.getText().toString();
    db.execSQL("INSERT INTO student VALUES('" + name + "')");
    showNames();
});

showNames(); // Show existing names
}

void showNames() {
    Cursor c = db.rawQuery("SELECT * FROM student", null);
    StringBuilder names = new StringBuilder();
    while (c.moveToNext()) names.append(c.getString(0)).append("\n");
    output.setText(names.toString());
}
}

```

Output:



Program11: Create an application to send SMS and receive SMS

Layout File: activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    android:id="@+id/main"
    android:layout_width="match_parent"
```

```
android:layout_height="match_parent"
android:orientation="vertical"
android:padding="20dp">
```

```
<EditText
    android:id="@+id/e1"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:hint="Enter Mobile Number"
    android:inputType="phone" />
```

```
<EditText
    android:id="@+id/e2"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:hint="Enter Message"
    android:inputType="textMultiLine" />
```

```
<Button
    android:id="@+id/btn"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Send SMS" />
```

```
</LinearLayout>
```

Java Code: MainActivity.java

```
package com.example.myapplication;
```

```
import android.content.Intent;
import android.net.Uri;
import android.os.Bundle;
import android.widget.Button;
import android.widget.EditText;
```

```
import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;
```

```
public class MainActivity extends AppCompatActivity {
```

```
    @Override
```

```

protected void onCreate(Bundle savedInstanceState) {
    super.onCreate(savedInstanceState);
    EdgeToEdge.enable(this);
    setContentView(R.layout.activity_main);
    ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main), (v,
insets) -> {
        Insets systemBars =
insets.getInsets(WindowInsetsCompat.Type.systemBars());
        v.setPadding(systemBars.left, systemBars.top, systemBars.right,
systemBars.bottom);
        return insets;
    });
}

```

```

EditText phone = findViewById(R.id.e1);
EditText message = findViewById(R.id.e2);
Button sendBtn = findViewById(R.id.btn);

```

```

sendBtn.setOnClickListener(v -> {
    String number = phone.getText().toString();
    String msg = message.getText().toString();

    Intent intent = new Intent(Intent.ACTION_VIEW);
    intent.setData(Uri.parse("sms:" + number));
    intent.putExtra("sms_body", msg);

```

```

        startActivity(intent);
    });
}

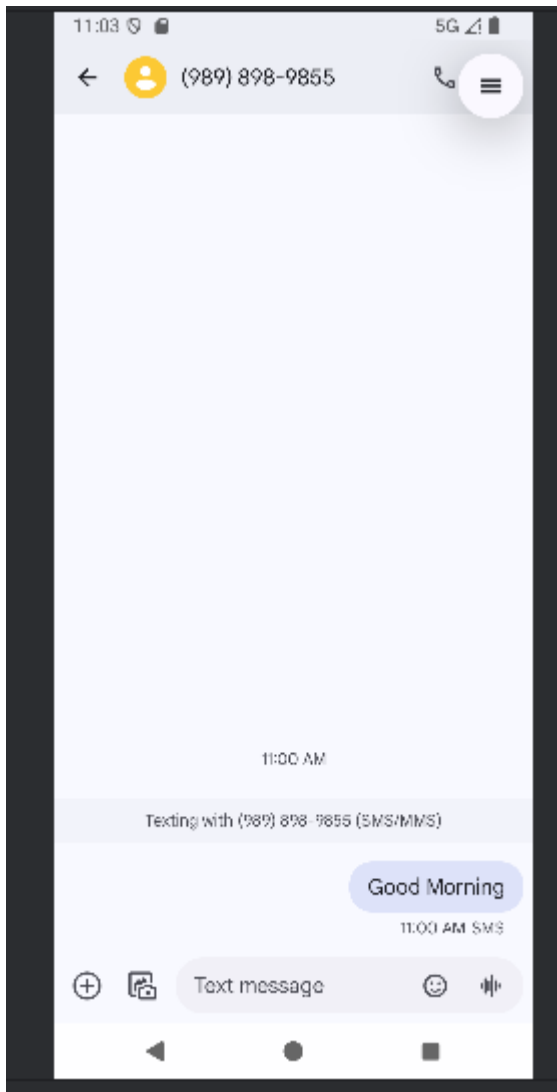
```

```

}

```

Output



Program12: Create an application to send an email

Layout File (activity_main.xml):

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">
```

```
    <LinearLayout
        android:layout_width="409dp"
        android:layout_height="729dp"
        android:orientation="vertical"
```

```
app:layout_constraintTop_toTopOf="parent"  
tools:layout_editor_absoluteX="1dp">
```

```
<EditText  
    android:id="@+id/et1"  
    android:layout_width="match_parent"  
    android:layout_height="wrap_content"  
    android:ems="10"  
    android:inputType="text"  
    android:text="Email Address" />
```

```
<EditText  
    android:id="@+id/et2"  
    android:layout_width="match_parent"  
    android:layout_height="wrap_content"  
    android:ems="10"  
    android:inputType="text"  
    android:text="Subject" />
```

```
<EditText  
    android:id="@+id/et3"  
    android:layout_width="match_parent"  
    android:layout_height="wrap_content"  
    android:ems="10"  
    android:inputType="text"
```



```

        android:text="Message" />

<Button
    android:id="@+id/btn"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="Send" />
</LinearLayout>
</androidx.constraintlayout.widget.ConstraintLayout>

```

Java File (MainActivity.java):

```

package com.example.emailapp;

import android.content.Intent;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;

import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_main);

        ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main), (v, insets) -> {
            Insets systemBars = insets.getInsets(WindowInsetsCompat.Type.systemBars());
            v.setPadding(systemBars.left, systemBars.top, systemBars.right, systemBars.bottom);
            return insets;
        });

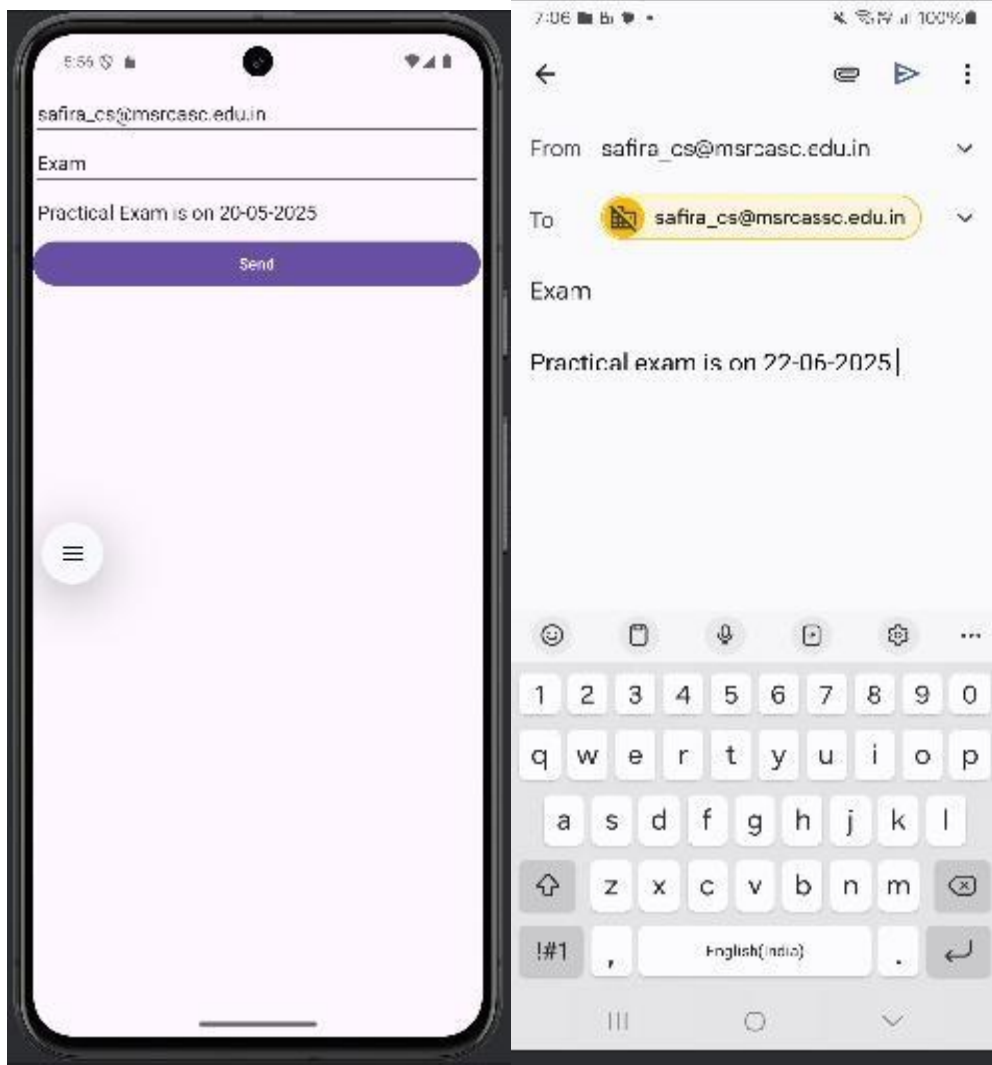
        EditText et1 = findViewById(R.id.et1);
        EditText et2 = findViewById(R.id.et2);
        EditText et3 = findViewById(R.id.et3);
    }
}

```

```
Button btn = findViewById(R.id.btn);

btn.setOnClickListener(view -> {
    String emailAddress = et1.getText().toString();
    String subject = et2.getText().toString();
    String message = et3.getText().toString();

    Intent intent = new Intent(Intent.ACTION_SEND);
    intent.setType("text/plain");
    intent.putExtra(Intent.EXTRA_EMAIL, new String[]{emailAddress});
    intent.putExtra(Intent.EXTRA_SUBJECT, subject);
    intent.putExtra(Intent.EXTRA_TEXT, message);
    startActivity(Intent.createChooser(intent, "Choose an Email app"));
});
}
}
```



Program 13: Display Map based on the current/given Location

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
```

```
xmlns:android="http://schemas.android.com/apk/res/android"
xmlns:app="http://schemas.android.com/apk/res-auto"
xmlns:tools="http://schemas.android.com/tools"
android:id="@+id/main"
android:layout_width="match_parent"
android:layout_height="match_parent"
tools:context=".MainActivity">
```

```
<Button
    android:id="@+id/btn1"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Google Map"
    tools:layout_editor_absoluteX="139dp"
    tools:layout_editor_absoluteY="209dp" />
```

```
</androidx.constraintlayout.widget.ConstraintLayout>
```

MainActivity.java

```
package com.example.myapplication;
```

```
import android.content.Intent;
import android.net.Uri;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
```

```
import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;
```

```
public class MainActivity extends AppCompatActivity {
```

```

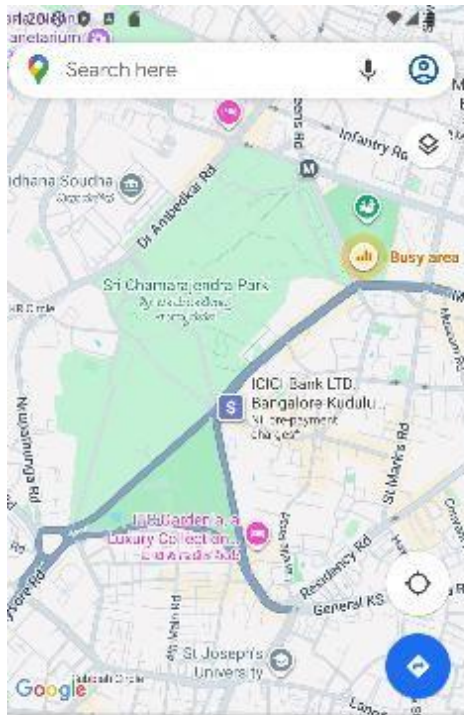
@Override
protected void onCreate(Bundle savedInstanceState) {
    super.onCreate(savedInstanceState);
    EdgeToEdge.enable(this);
    setContentView(R.layout.activity_main);
    ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main), (v, insets) -> {
        Insets systemBars = insets.getInsets(WindowInsetsCompat.Type.systemBars());
        v.setPadding(systemBars.left, systemBars.top, systemBars.right, systemBars.bottom);
        return insets;
    });

    Button b1=findViewById(R.id.btn1);
    b1.setOnClickListener(new View.OnClickListener() {
        @Override
        public void onClick(View v) {
            Intent i = new
Intent(Intent.ACTION_VIEW,Uri.parse("geo:12.9716,77.5946"));
            startActivity(i);
        }
    });

}
}

```

Output



Latest in Ashok Nagar

 **Ali Al romaithi**
Local Guide · 124 photos



    
Explore Go Saved Contribute Updates